

Abduxoliq Ashuraliyev
Independent Researcher
Tashkent, Uzbekistan

May 11, 2026

Editorial Office
Nature Sustainability
Springer Nature

Dear Editor,

I submit the manuscript “**Governance Reform Unlocks Decarbonization Dead Zones in Public Procurement**” for consideration as an Article in *Nature Sustainability*.

Summary

This manuscript identifies a climate-implementation bottleneck in public procurement: green criteria are least effective where governments lack credible supplier choice. Using 21.6 million government contracts from 27 countries, it shows that 22 “Decarbonization Dead Zones” place an estimated €190–250 billion of annual EU spending in markets where high carbon intensity coincides with persistent single-bidder awards.

This submission contributes three linked findings:

- In EU-context countries, competitive contracts are more common in higher-carbon sectors, yielding a -4.3% allocative portfolio premium ($t = -110$; $d = -0.08$).
- Directive 2014/24/EU is associated with a -7.2 percentage-point reduction in single-bidder rates in the preferred staggered Callaway & Sant’Anna design (RMSPE permutation $p = 0.042$); leave-one-out bounds remain -4.75 to -8.46 pp, and an external CanadaBuys procurement-method proxy-control gives -3.6 pp ($p = 0.0068$).
- The allocative channel is corroborated across independent facility-, sector-, and supplier-level evidence, including a 43% within-sector EU ETS emissions gap, a 66% post-reform narrowing in Eurostat, annual E-PRTR facility matching, negative within-supplier comparisons, and a roughly fivefold increase in access to SBTi-validated suppliers when moving from one to five bidders in Dead Zone sectors.

We believe this contribution fits *Nature Sustainability* because it identifies a missing institutional precondition for climate-effective procurement: before governments can use GPP to select cleaner suppliers, they must first open carbon-intensive markets to competition. The paper therefore presents governance reform not as administrative clean-up, but as an implementable climate-policy lever with immediate relevance at the scale of €190–250 billion in annual EU procurement spending.

Reproducibility

All data are publicly available:

- EU Tenders Electronic Daily: <https://ted.europa.eu>
- EXIOBASE 3.8.2: <https://doi.org/10.5281/zenodo.5589597>
- Harmonised dataset and repository-aligned Data.zip archive (21.6M contracts): Zenodo DOI:

10.5281/zenodo.20098951

- Analysis code: <https://github.com/ProgrmerJack/Public-Procurement-Control-Surface>

Suggested Reviewers

- **Prof. Mihály Fazekas**, Department of Public Policy, Central European University.
FazekasM@ceu.edu — procurement analytics, single-bidding, and corruption measurement.
- **Prof. Erica L. Plambeck**, Operations, Information & Technology, Stanford Graduate School of Business.
elp@stanford.edu — sustainable supply chains and environmental operations.
- **Dr. Konstantin Stadler**, Industrial Ecology Digital Laboratory, Department of Energy and Process Engineering, NTNU.
konstantin.stadler@ntnu.no — EXIOBASE, multi-regional input-output analysis, and carbon accounting. We flag this as an optional suggestion because he contributed to the EXIOBASE data infrastructure used in this manuscript.

Declarations

- The manuscript has not been published elsewhere and is not under consideration at another journal.
- The author declares no competing interests.
- No human subjects or animal research was involved.
- All data are publicly available administrative records.

Thank you for considering this submission.

Sincerely,

Abduxoliq Ashuraliyev
Independent Researcher
Tashkent, Uzbekistan
jack00040008@outlook.com
ORCID: 0009-0003-5482-5526